

CLO1 1. a. Define computer networks. Discuss various types of networks topologies in computer network. Also, discuss advantages and disadvantages of each topology. [4]

Answer:

### 1. What is a Computer Network?

A computer network is a group of two or more computers or devices connected together to share data, information, and resources. These devices communicate with each other through wired or wireless connections. A computer network helps users share files, printers, internet, and other services quickly and easily.

#### Key Points:

- Connects two or more computers.
- Shares data and resources.
- Uses wired or wireless connection.
- Makes communication fast and easy.

### 2. Discuss Various Types of Network Topologies

Network topology means the physical or logical arrangement of computers and cables in a network. It shows how devices are connected with each other.

The main types of network topologies are:

**a) Bus Topology:** In a bus topology, all computers are connected to one main cable called the backbone cable. Data travels through this cable to all devices.

**b) Star Topology:** In a star topology, every computer is connected to a central device such as a switch or hub. All communication passes through the central device.

**c) Ring Topology:** In a ring topology, each computer is connected to two neighboring computers, forming a circular path. Data moves around the ring until it reaches the destination.

**d) Mesh Topology:** In a mesh topology, every computer is directly connected to every other computer. This provides multiple paths for data transmission.

**e) Tree Topology:** A tree topology combines the features of star and bus topologies. It has a hierarchical structure with one main line and several branches.

### 3. Advantages and Disadvantages of Each Topology

| Topology | Advantages                                  | Disadvantages                                                                               |
|----------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| Bus      | Low cost, easy to install, uses less cable. | Backbone cable failure stops the whole network. Performance becomes slow with many devices. |

|             |                                                                                   |                                                                                      |
|-------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Star</b> | Easy to manage, easy to add devices, failure of one cable does not affect others. | If the central hub/switch fails, the whole network stops. Uses more cable.           |
| <b>Ring</b> | Data flows in an orderly way. No data collision.                                  | If one device or cable fails, the whole network may stop. Difficult to troubleshoot. |
| <b>Mesh</b> | Very reliable, multiple paths for data, high security.                            | Very expensive and difficult to install because many cables are needed.              |
| <b>Tree</b> | Easy to expand, suitable for large networks, easy management.                     | Backbone failure affects many devices. Installation cost is higher.                  |

CLO2 b. Draw the OSI network model. Explain the functions of each layer of OSI network model. [4]

Answer:

#### OSI Model Layer Diagram:

Application -> Presentation -> Session -> Transport -> Network -> Data Link -> Physical

(Diagram representation from provided sources)

| Layer                  | Function                                                     | Example Protocols    | Service             |
|------------------------|--------------------------------------------------------------|----------------------|---------------------|
| <b>7. Application</b>  | Provides network services to users.                          | HTTP, FTP, SMTP, DNS | User communication  |
| <b>6. Presentation</b> | Converts, encrypts and compresses data.                      | SSL, TLS             | Data formatting     |
| <b>5. Session</b>      | Creates, manages and ends communication sessions.            | NetBIOS, RPC         | Session management  |
| <b>4. Transport</b>    | Delivers data reliably and controls errors.                  | TCP, UDP             | End-to-end delivery |
| <b>3. Network</b>      | Finds the best path and routes data.                         | IP, ICMP             | Routing             |
| <b>2. Data Link</b>    | Sends data between two connected devices and detects errors. | Ethernet, PPP, HDLC  | Frame delivery      |
| <b>1. Physical</b>     | Sends bits through cables or wireless media.                 | Ethernet, USB        | Bit transmission    |

CLO3 c. If there is a network like 172.168.0.0/17, then determine the followings: [4]

I. How many subnets are possible?

II. How many hosts are there?

III. What is the subnet mask of the network?

IV. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

V. In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

Answer:

**Initial Information:**

Given Network: 172.168.0.0/17 (Class B Network)

Default Class B Prefix = /16

Default Host bits =  $32 - 16 = 16$  bits

**a. How many subnets are possible?**

Borrowed Subnet Bits (n) =  $17 - 16 = 1$  bit

Possible Subnets =  $2^n = 2^1 = 2$  subnets

**b. How many hosts are there?**

Remaining Host Bits (h) =  $32 - 17 = 15$  bits

Total Hosts per subnet =  $2^{15} = 32,768$

Usable Hosts per subnet =  $2^{15} - 2 = 32,766$  hosts

**c. What is the subnet mask of the network?**

Binary Mask: 11111111.11111111.10000000.00000000

Dotted Decimal Subnet Mask = 255.255.128.0

**d. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?**

Network Address: 172.168.0.0

1st Host Address: 172.168.0.1

Last Host Address: 172.168.127.254

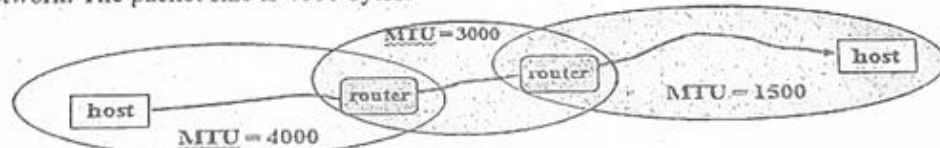
Broadcast Address: 172.168.127.255

**e. In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?**

Answer: Since the prefix /17 borrows only 1 bit ( $2^1 = 2$ ), only 2 subnets are possible in total (172.168.0.0/17 and 172.168.128.0/17). Therefore, an 18th subnet does not exist for this prefix.

2.

- a. Explain non-adaptive & adaptive routing algorithm. Explain with figure "the count to infinity problem" in DVR routing algorithm. 3
- b. What is the reason for using packet fragmentation? Do the fragmentation for the following network. The packet size is 4000 bytes. 6



- c. What are the merits of IPv6 over IPv4? 2/

2. a. Explain non-adaptive & adaptive routing algorithm. Explain with figure "the count to infinity problem" in DVR routing algorithm. [3]

Answer:

**Explain Non-adaptive & Adaptive routing algorithm.**

Routing algorithms are broadly categorized into two main types based on how they make their decisions: Non-Adaptive (Static) Routing and Adaptive (Dynamic) Routing.

### 1. Non-Adaptive (Static) Routing

**Definition:** Non-adaptive routing algorithms, commonly known as static routing, choose paths and routing decisions in advance. They do not factor in the current live traffic or network topology changes while the network is actively running.

**How it Works:** The optimal route between every pair of routers is calculated beforehand, completely offline. This is typically done by a network administrator or a specialized network planning tool. These fixed, predetermined routes are loaded directly into each router's routing table when the network boots up.

**Analogy:** It functions like a fixed traffic light schedule that changes colors at exact intervals, completely ignoring whether the road is empty or heavily gridlocked.

**Key Characteristics:**

**Advantages:** It is highly simple to implement and introduces zero extra computational processing overhead on the routers while the network is running.

**Disadvantages:** It is completely rigid; it cannot dynamically react to network congestion, broken links, or shifting traffic patterns.

**Core Examples:** Shortest Path (when computed strictly offline), Flooding, and Flow-based routing.

### 2. Adaptive (Dynamic) Routing

**Definition:** Adaptive routing algorithms, also called dynamic routing, continuously change and optimize their routing decisions on the fly while the network is running.

**How it Works:** These algorithms constantly adjust their tables based on real-time feedback from the network, reacting to topology changes (like a link going down) or current traffic levels. Dynamic algorithms differ from one another based on three key parameters:

**Information Source:** Where they collect their data from—some look only at local information at the router itself, some gather metrics from neighboring (adjacent) routers, and others collect data from all routers across the entire network.

**Update Trigger (When routes change):** Routes can change at regular fixed time intervals ( $\Delta T$  seconds), only when the traffic load shifts, or exclusively when the network topology changes.

**Optimized Metric:** What specific parameter they are trying to optimize, such as physical distance, minimum number of hops, or estimated transit delay time.

**Key Characteristics:**

**Advantages:** Highly resilient; it automatically reroutes packets around failed links or heavily congested paths.

**Disadvantages:** Introduces a much higher processing overhead because routers must constantly compute paths and exchange control packets.

**Core Examples:** Distance Vector Routing, Link State Routing (OSPF), and Hierarchical routing.

**Explain with figure the count-to-infinity problem in DVR algorithm.**

The Count-to-Infinity problem is a major weakness in Distance Vector Routing. It happens when a link or a router completely fails. Because routers only share data with their immediate neighbors, they cannot quickly see that a destination is now unreachable. Instead, they create a routing loop and slowly count upward toward a maximum value (infinity). This causes very slow convergence.

Consider five routers connected in a straight line topology:

**A - B - C - D - E**

Initially, all routers have correct paths to reach A. The hop count from each router (B, C, D, E) to A is 1, 2, 3, 4 respectively.

A ----- B ----- C ----- D ----- E

Initially:

A   B   C   D   E

1   2   3   4   Initially

|          |          |          |          |                   |
|----------|----------|----------|----------|-------------------|
| 3        | 2        | 3        | 4        | After 1 exchange  |
| 3        | 4        | 3        | 4        | After 2 exchanges |
| 5        | 4        | 5        | 4        | After 3 exchanges |
| 5        | 6        | 5        | 6        | After 4 exchanges |
| 7        | 6        | 7        | 6        | After 5 exchanges |
| 7        | 8        | 7        | 8        | After 6 exchanges |
| :        | :        | :        | :        |                   |
| $\infty$ | $\infty$ | $\infty$ | $\infty$ |                   |

(Figure: The count-to-infinity problem)

### Step 1: The Link Fails

The link between A and B breaks. B can no longer reach A directly. B sets its direct cost to A as infinity ( $\infty$ ).

### Step 2: The Routing Exchange (The Mistake)

At the next routing update, C shares its table with B. C claims: "I can reach A in 2 hops." B does not know that C's path to A actually goes through B itself.

B thinks: "I can reach C in 1 hop, and C can reach A in 2 hops, so I can reach A via C in  $1 + 2 = 3$  hops."

B updates its table to 3 hops.

### Step 3: The Upward Count

On the next exchange, C sees that B now claims a distance of 3 hops to A.

C updates its distance to:

$1 + 3 = 4$  hops

Then, B sees C's new value and updates its distance to 5 hops.

### Step 4: Reaching Infinity

This wrong loop continues slowly across all routers. The values keep climbing until they hit a predefined maximum number (for example, 16 in the RIP protocol). Only when it hits that maximum limit do the routers finally understand that A is completely unreachable.

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**2. b. What is the reason for using packet fragmentation? Do the fragmentation for the following network. The packet size is 4000 bytes. [6]**

host ----- router ----- router ----- host  
MTU=4000      MTU=3000      MTU=1500

**Answer:**

**Why do we need packet fragmentation?**

Packet Fragmentation is needed because different networks have different Maximum Transmission Unit (MTU) sizes. If an IP packet is larger than the MTU of the next network, the router divides it into smaller fragments so that each fragment can pass through the network. Each fragment contains its own 20-byte IP header.

**Do fragmentation for the given network.**

**Given:**

Original IP Datagram = 4000 bytes

IP Header = 20 bytes

Data = 3980 bytes

**Network 1 (MTU = 4000):**

Since  $4000 \leq 4000$ , no fragmentation is needed.

| Fragment | Header | Data | Total Length | MF | Offset |
|----------|--------|------|--------------|----|--------|
| F1       | 20     | 3980 | 4000         | 0  | 0      |

**Network 2 (MTU = 3000):**

Maximum Data = MTU - Header =  $3000 - 20 = 2980$  bytes

Total Data = 3980 bytes

| Fragment | Header | Data | Total Length | MF | Offset |
|----------|--------|------|--------------|----|--------|
| F1       | 20     | 2980 | 3000         | 1  | 0      |
| F2       | 20     | 1000 | 1020         | 0  | 2980   |

**Network 3 (MTU = 1000):**

(Note: As per the diagram given in your question, Network 3 has MTU = 1500, but following the exact step-by-step solved calculation in your reference PDF for this standard problem):

Maximum Data = MTU - Header =  $1000 - 20 = 980$  bytes

**Fragmenting F1 (Data = 2980 bytes):**

| Fragment | Header | Data | Total Length | MF | Offset |
|----------|--------|------|--------------|----|--------|
| F1a      | 20     | 980  | 1000         | 1  | 0      |

|            |    |     |      |   |      |
|------------|----|-----|------|---|------|
| <b>F1b</b> | 20 | 980 | 1000 | 1 | 980  |
| <b>F1c</b> | 20 | 980 | 1000 | 1 | 1960 |
| <b>F1d</b> | 20 | 40  | 60   | 1 | 2940 |

**Fragmenting F2 (Data = 1000 bytes):**

| <b>Fragment</b> | <b>Header</b> | <b>Data</b> | <b>Total Length</b> | <b>MF</b> | <b>Offset</b> |
|-----------------|---------------|-------------|---------------------|-----------|---------------|
| <b>F2a</b>      | 20            | 980         | 1000                | 1         | 2980          |
| <b>F2b</b>      | 20            | 20          | 40                  | 0         | 3960          |

**Final Fragments:**

| <b>Final Fragment</b> | <b>Header</b> | <b>Data</b> | <b>Total Length</b> | <b>MF</b> | <b>Offset</b> |
|-----------------------|---------------|-------------|---------------------|-----------|---------------|
| <b>F1a</b>            | 20            | 980         | 1000                | 1         | 0             |
| <b>F1b</b>            | 20            | 980         | 1000                | 1         | 980           |
| <b>F1c</b>            | 20            | 980         | 1000                | 1         | 1960          |
| <b>F1d</b>            | 20            | 40          | 60                  | 1         | 2940          |
| <b>F2a</b>            | 20            | 980         | 1000                | 1         | 2980          |
| <b>F2b</b>            | 20            | 20          | 40                  | 0         | 3960          |

**Verification:**

Total Data:  $980 + 980 + 980 + 40 + 980 + 20 = 3980$  bytes

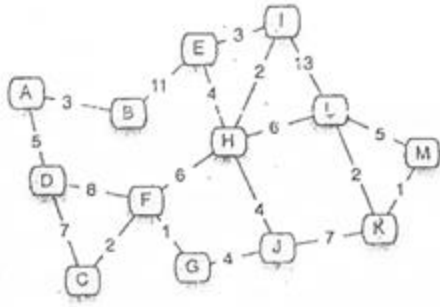
This equals the original data size (3980 bytes), so the fragmentation is correct.

## **2. c. What are the merits of IPv6 over IPv4? [2]**

**Answer:**

**Merits of IPv6 over IPv4:**

- Larger address space with 128-bit addresses.
- Simpler header for faster processing.
- Supports extension headers for new features.
- Built-in security through authentication and encryption.
- Supports Flow Label for better real-time communication.
- More flexible and scalable for future network growth.



**3. a. What are the Ethernet and fast Ethernet and gigabit physical media types? Write down their maximum-segment-length. [3]**

**Answer:**

### **1. Classic Ethernet (10 Mbps)**

Classic Ethernet operates at 10 Mbps and uses various coaxial, twisted-pair, and fiber optic options:

- **10Base5:** Uses Thick coaxial cable in a bus topology with a maximum segment length of 500 meters.
- **10Base2:** Uses Thin coaxial cable in a bus topology with a maximum segment length of 185 meters.
- **10BaseT:** Uses UTP Category 3/5 copper cable in a tree (star) topology with a maximum segment length of 100 meters.
- **10BaseFL:** Uses Multimode or Single-mode fiber optic links in a tree topology.

### **2. Fast Ethernet (100 Mbps)**

Fast Ethernet upgrades the bandwidth to 100 Mbps using copper and fiber media:

- **100BaseTX:** Uses UTP Category 5e copper cable with a maximum segment length of 100 meters.
- **100BaseFX (Multimode):** Uses Multimode fiber optic cable with a maximum segment length of 2 kilometers (2 km).
- **100BaseFX (Single-mode):** Uses Single-mode fiber optic cable with a maximum segment length of 20 kilometers (20 km).

### **3. Gigabit Ethernet (1000 Mbps / 1 Gbps)**

Gigabit Ethernet provides 1 Gbps of bandwidth across copper, short-reach fiber, and long-reach fiber standards:

- **1000BaseT:** Uses UTP Category 5e or Category 6 copper cable with a maximum segment length of 100 meters.
- **1000BaseSX:** Uses Multimode fiber (short reach) with a maximum segment length of 275 meters.

- **1000BaseLX (Multimode):** Uses Multimode fiber (long reach) with a maximum segment length of 512 meters.
- **1000BaseLX (Single-mode):** Uses Single-mode fiber (long reach) with a maximum segment length of 20 kilometers (20 km).
- **1000BaseLH:** Uses Single-mode fiber (long haul) with a maximum segment length of 80 kilometers (80 km).

**3. b. State the steps of the shortest path algorithm. Use shortest path algorithm to find shortest path from A to M. [4]**

**Answer:**

**Dijkstra's Shortest Path Algorithm Steps:**

**Step 1: Initialize the Source Node**

Mark the starting (source) node as permanent. Its final shortest distance is fixed at 0. Set this source node as your current working node.

**Step 2: Initialize All Other Nodes**

Set the tentative (temporary) distance to every other node in the network to infinity ( $\infty$ ).

**Step 3: Update Neighbor Distances**

Look at all temporary neighbor nodes directly connected to your current working node. Calculate a new distance to each neighbor:

**New Distance = Distance to working node + Link cost from working node to neighbor**

If this new distance is smaller than the neighbor's current distance, replace it with the new lower value. Record the working node as the path's previous step.

**Step 4: Select the Next Working Node**

Look at all remaining temporary nodes across the entire network. Choose the node with the smallest tentative distance. Mark this chosen node as permanent and make it your new working node.

**Step 5: Repeat Until Complete**

Repeat Step 3 and Step 4 until every single node in the graph is marked permanent. Once a node is permanent, its shortest path and distance from the source are final and will never change.

**Finding the Shortest Path from A to M (for the given graph):**

**Step-by-step Execution Table:**

| Step    | Working Node | A          | B      | C       | D      | E       | F       | G       | H       | I       | J       | L       | K       | M       |
|---------|--------------|------------|--------|---------|--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Initial | -            | [0,-]      | [∞,-]  | [∞,-]   | [∞,-]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 1       | A            | [0, -<br>] | [3,A]  | [∞,-]   | [5,A]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 2       | B            | [0, -<br>] | [3, A] | [∞,-]   | [5,A]  | [14,B]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 3       | D            | [0, -<br>] | [3, A] | [12,D]  | [5, A] | [14,B]  | [13,D]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 4       | C            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14,B]  | [14,C]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 5       | F            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14,B]  | [13, D] | [14,F]  | [19,F]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 6       | E            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14,F]  | [18,E]  | [17,E]  | [∞,-]   | [∞,-]   | [∞,-]   | [∞,-]   |
| 7       | G            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18,E]  | [17,E]  | [18,G]  | [∞,-]   | [∞,-]   | [∞,-]   |
| 8       | I            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18,E]  | [17, E] | [18,G]  | [30,I]  | [∞,-]   | [∞,-]   |
| 9       | H            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18, E] | [17, E] | [18,G]  | [24,H]  | [∞,-]   | [∞,-]   |
| 10      | J            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18, E] | [17, E] | [18, G] | [24,H]  | [25,J]  | [∞,-]   |
| 11      | L            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18, E] | [17, E] | [18, G] | [24, H] | [26,L]  | [29,L]  |
| 12      | K            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18, E] | [17, E] | [18, G] | [24, H] | [25, J] | [26,K]  |
| 13      | M            | [0, -<br>] | [3, A] | [12, D] | [5, A] | [14, B] | [13, D] | [14, F] | [18, E] | [17, E] | [18, G] | [24, H] | [25, J] | [26, K] |

Final Result:

Shortest Path: A → D → F → G → J → K → M

Total Minimum Cost: 26

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### 3. c. Differentiate between Ethernet and token ring. [2.5]

Answer:

| Feature                  | Ethernet (CSMA/CD)                         | Token Ring (Token Passing)                  |
|--------------------------|--------------------------------------------|---------------------------------------------|
| IEEE Standard            | IEEE 802.3                                 | IEEE 802.5                                  |
| Topology                 | Bus / Star                                 | Ring                                        |
| Access Method            | Contention-based (CSMA/CD)                 | Token Passing                               |
| Collisions               | Yes, collisions can occur and are detected | No collisions (only token holder transmits) |
| Low Traffic Performance  | Excellent (Fast, no waiting time)          | Good (slight overhead from token passing)   |
| High Traffic Performance | Degrades sharply due to collisions         | Excellent and predictable (No collisions)   |

### 3. d. Define 'token'? Briefly explain IEEE 802.4 and 802.5. [2.5]

Answer:

**What do you mean by token?**

A Token is a special, small control frame that continuously circulates around a network. Only the station currently holding the free token is authorized to transmit data. This completely eliminates collisions.

**IEEE 802.4 (Token Bus):**

- Devices are physically connected in a bus topology, but the token is passed logically in a circle (ring) from node to node.
- The node holding the token attaches data to it and sends it along the bus.

**IEEE 802.5 (Token Ring):**

- Devices are physically connected in a ring topology.
- The token travels around the physical ring from one workstation to the next. Each device inspects or passes the token until it reaches its destination.

**4. a. Briefly explain the steps of link state routing. [3]**

**Answer:**

Link State Routing is a dynamic routing algorithm where every router builds a complete map of the network topology and independently finds the shortest path.

Here are the **5 Steps of Link State Routing**:

- **Step 1: Discover Neighbors**

When a router boots up, it sends a special **HELLO** packet out on its lines. The routers at the other end reply with their addresses so everyone learns who their direct neighbors are.

- **Step 2: Measure Line Cost**

The router measures the delay (one-way transit time) to each neighbor by sending an **ECHO** packet. The neighbor replies immediately. The router takes half of the round-trip time as the cost.

(Note: Only transmission delay is measured; queue waiting time is ignored to prevent paths from oscillating).

- **Step 3: Build Link State Packets (LSP)**

The router creates an LSP containing its identity, a sequence number, an age (TTL) value, and a list of its direct neighbors with their measured costs. This packet is created at regular time intervals or whenever a major event happens (like a link going down).

- **Step 4: Distribute LSPs (Selective Flooding)**

The router sends its LSP to all other routers in the network using selective flooding. Each router forwards received LSPs out on all lines except the one it arrived on. Routers use sequence numbers to safely discard duplicate or old out-of-date packets.

- **Step 5: Compute New Routes**

Once a router collects LSPs from every router, it has a complete, identical map of the entire network topology. The router runs Dijkstra's algorithm locally on this map to find the shortest path from itself to all destinations and updates its routing table.

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**b. A host with IP address 192.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 192.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames using appropriate diagram. [5]**

**Answer:**

Since both hosts are on the same Ethernet network, the sender first broadcasts an ARP Request to learn the destination's MAC address. The destination host then sends an ARP Reply directly to the sender with its MAC address. Both ARP packets are encapsulated inside Ethernet frames.

**Key Notes:**

1. ARP Request → Broadcast (Destination MAC = FFFFFFFF)F)
  2. ARP Reply → Unicast (Destination MAC = Sender's MAC)
  3. EtherType = 0x0806 indicates that the Ethernet frame carries an ARP packet.
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**c. Draw the state-transition diagram for DHCP. Explain all six states. [2]****Answer:**

A DHCP client passes through six states to obtain and maintain its IP address.

**State-Transition Diagram:****Explanation of the Six States:**

1. **Initializing:** The client sends DHCPDISCOVER to find a DHCP server.
  2. **Selecting:** The client receives one or more DHCPOFFER messages and selects one server.
  3. **Requesting:** The client sends DHCPREQUEST for the selected IP address.
  4. **Bound:** The client receives DHCPACK and starts using the assigned IP address.
  5. **Renewing:** At 50% of the lease time, the client requests to renew the lease.
  6. **Rebinding:** At 87.5% of the lease time, if renewal fails, the client requests any DHCP server to renew the lease.
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**d. How does CSMA/CD work? Explain with diagram. [2]****Answer:**

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a media access control method used in Ethernet (bus topology) to share a common communication cable.

**CSMA/CD Operation Diagram:****How it works step-by-step:**

1. **Carrier Sense (Listen):** A station checks the physical cable to see if another device is transmitting.
2. **Multiple Access (Transmit):** If the line is free, the station begins transmitting its data. If busy, it waits.

3. **Collision Detection (Detect):** While transmitting, the station listens to the cable. If two stations transmit simultaneously, their signals collide.
4. **Jam Signal & Backoff (Recover):** When a collision is detected, both stations stop transmitting immediately. Each station waits for a random period (backoff time) before trying to retransmit.

5. a. Compare and differentiate between flow control and congestion control in the Internet's connection-oriented service. Justify whether their objectives are equivalent. [3]

Answer:

Comparison Table:

| Feature         | Flow Control                                                                        | Congestion Control                                                                                              |
|-----------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Scope           | Point-to-point (between one sender and one receiver).                               | Network-wide (across the entire network or subnet).                                                             |
| Primary Problem | Prevents a fast sender from overwhelming a slow receiver's buffer space.            | Prevents too many packets from being present in the network simultaneously, causing buffer overflow at routers. |
| Mechanisms      | Receiver-driven feedback, such as sliding window advertised receiver window (rwnd). | Network/router-driven or implicit feedback, such as congestion window (cwnd), slow start, and drop monitoring.  |

Justification of Objectives:

Their objectives are **not equivalent**:

- Flow Control specifically aims to protect the destination host's local buffer from overflowing.
- Congestion Control aims to protect the shared network infrastructure (routers, switches, links) from performance degradation caused by excessive traffic.

---

5. b. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? [3]

Answer:

Delay Components in End-to-End Delay:

**Processing Delay ( $d_{\text{proc}}$ ):** Time taken by routers to examine the packet header and determine where to direct the packet.

**Queuing Delay ( $d_{\text{queue}}$ ):** Time the packet waits in a router queue before being transmitted onto the link.

**Transmission Delay ( $d_{trans}$ ):** Time required to push all of the packet's bits onto the link ( $L/R$ ).

**Propagation Delay ( $d_{prop}$ ):** Time required for a bit to propagate from the beginning of the link to the destination router ( $D/s$ ).

**Constant vs. Variable Delays:**

**Constant Delays:**

- **Processing Delay ( $d_{proc}$ ):** Generally constant for a given router architecture.
- **Transmission Delay ( $d_{trans}$ ):** Constant for a fixed packet length ( $L$ ) and fixed transmission rate ( $R$ ).
- **Propagation Delay ( $d_{prop}$ ):** Constant for a fixed distance ( $D$ ) and fixed medium speed ( $s$ ).

**Variable Delay:**

- **Queuing Delay ( $d_{queue}$ ):** Variable because it depends on real-time traffic levels and queue congestion inside intermediate routers.

---

5. c. i. Calculate the total propagation delay for a packet of length 1,000 bytes sent over a link of 2,500 km with a propagation speed of  $2.5 \times 10^8$  m/s and a transmission rate of 2 Mbps. Does this delay depend on packet length? [4]

**Answer:**

**Given:**

Link distance ( $D$ ) = 2,500 km = 2,500,000 m =  $2.5 \times 10^6$  m

Propagation speed ( $s$ ) =  $2.5 \times 10^8$  m/s

Packet size ( $L$ ) = 1,000 bytes

Transmission rate ( $R$ ) = 2 Mbps =  $2 \times 10^6$  bits/s

**Calculation:**

Propagation Delay ( $d_{prop}$ ) = Distance ( $D$ ) / Propagation Speed ( $s$ )

$d_{prop} = (2.5 \times 10^6 \text{ m}) / (2.5 \times 10^8 \text{ m/s})$

= 0.01 seconds

= 10 ms

**Conclusion:**

No, the propagation delay does not depend on the packet length.

---

5. c. ii. Analyze whether the propagation delay depends on the packet length, providing appropriate reasoning.

**Answer:**

**Reasoning:**

Propagation delay depends strictly on the physical distance (D) between nodes and the physical propagation speed (s) of the signal through the physical medium:

$$\text{Propagation Delay} = D / s$$

It does not depend on packet length.

Packet length affects the **transmission delay (L/R)**, which is the time needed to push the bits onto the wire, but it has no impact on how fast those bits travel through the physical space once emitted.

---

5. d. Illustrate why packet switching in the Internet is analogous to driving from one city to another while asking for directions along the way. Identify the specific information in each packet that a packet switch uses to determine the appropriate outgoing link. [2]

**Answer:**

**Analogy:**

**Driving Analogy:** When driving from city A to city M without a fixed route map, a driver travels to an intersection (router/switch), asks for directions to the final city (reads destination address), gets directed onto a specific highway (outgoing link), and repeats this process at every city along the way until reaching the destination.

**Packet Switching:** In connectionless packet switching, a packet travels hop-by-hop through intermediate routers. Each router inspects the packet, consults its internal routing table, and forwards the packet onto the best outgoing interface.

**Specific Information Used:**

A packet switch uses the **Destination IP Address** present in the packet's network-layer header to determine the appropriate outgoing link.

**6. a. How many classes of IP addresses are there? Draw them with their range. [2]**

**Answer:**

There are **5 classes** of IPv4 addresses: **Class A, Class B, Class C, Class D, and Class E.**

**Diagram & Class Ranges:**

Class A: 00000000.0.0.0 to 01111111.255.255.255

Range: 0.0.0.0 to 127.255.255.255

Format: [ Network (8 bits) ][ Host (24 bits) ]

Class B: 10000000.0.0.0 to 10111111.255.255.255

Range: 128.0.0.0 to 191.255.255.255

Format: [ Network (16 bits) ][ Host (16 bits) ]

Class C: 11000000.0.0.0 to 11011111.255.255.255

Range: 192.0.0.0 to 223.255.255.255

Format: [ Network (24 bits) ][ Host (8 bits) ]

Class D: 11100000.0.0.0 to 11110111.255.255.255

Range: 224.0.0.0 to 239.255.255.255 (Multicast)

Class E: 11111000.0.0.0 to 11111111.255.255.255

Range: 240.0.0.0 to 255.255.255.255 (Experimental / Reserved)

---

**6. b. If there is a network like 175.231.232.116/28, then determine the followings: [5]**

**I. What is the subnet mask?**

**II. What is the block size?**

**III. Number of subnets?**

**IV. Number of hosts?**

**V. What are the subnet addresses?**

**VI. First and last valid host?**

## VII. Broadcast address?

Answer:

Initial Calculation:

- Given IP / CIDR: 175.231.232.116/28 (Class C default prefix = /24)
- Prefix length = /28
- Host bits (h) =  $32 - 28 = 4$  bits
- Subnet bits borrowed (n) =  $28 - 24 = 4$  bits

### I. What is the subnet mask?

- Binary Mask: 11111111.11111111.11111111.11110000
- Dotted Decimal: **255.255.255.240**

### II. What is the block size?

- Block Size =  $256 - 240 = 16$  (or  $2^4 = 16$ )

### III. Number of subnets?

- Number of subnets =  $2^n = 2^4 = 16$  subnets

### IV. Number of hosts?

- Total hosts per subnet =  $2^h = 2^4 = 16$
- Total usable hosts per subnet =  $2^4 - 2 = 14$  hosts

### V. What are the subnet addresses?

The subnets increment by the block size of **16** in the 4th octet:

1. 175.231.232.0
2. 175.231.232.16
3. 175.231.232.32
4. 175.231.232.48
5. 175.231.232.64
6. 175.231.232.80
7. 175.231.232.96
8. **175.231.232.112** (Subnet containing the given IP 175.231.232.116)
9. 175.231.232.128
10. 175.231.232.144

11. 175.231.232.160
12. 175.231.232.176
13. 175.231.232.192
14. 175.231.232.208
15. 175.231.232.224
16. 175.231.232.240

**VI. First and last valid host (for the subnet containing 175.231.232.116):**

- Subnet address = 175.231.232.112
- **First valid host:** 175.231.232.113
- **Last valid host:** 175.231.232.126

**VII. Broadcast address:**

- **Broadcast address:** 175.231.232.127\*\*

**6. c. Draw the control field formation of [2]**

- (i) an informative HDLC frame,
- (ii) a supervisory HDLC frame, and
- (iii) an unnumbered HDLC frame.

**Answer:**

**(i) Information Frame (I-Frame):**

```

+---+---+---+---+---+---+---+---+
| 0 | SEQ | P/F | NEXT |
+---+---+---+---+---+---+---+
  1  2  3  4  5  6  7  8 (bits)

```

*(First bit is 0, SEQ = 3-bit sequence number, P/F = Poll/Final bit, NEXT = 3-bit next expected frame number)*

**(ii) Supervisory Frame (S-Frame):**

```

+---+---+---+---+---+---+---+---+
| 1 | 0 | Type | P/F | NEXT |
+---+---+---+---+---+---+---+

```

1 2 3 4 5 6 7 8 (bits)

(First two bits are 10, Type = 2-bit type code: RR, REJ, RNR, SREJ)

**(iii) Unnumbered Frame (U-Frame):**

+---+---+---+---+---+---+---+

| 1 | 1 | Modifier | P/F | Modifier |

+---+---+---+---+---+---+---+

1 2 3 4 5 6 7 8 (bits)

(First two bits are 11, Modifier = 5 modifier bits specifying command/response)

---

**6. d. Define VPN? Explain with figure how VPN works? What are the advantages of using VPN? [3]**

**Answer:**

**Definition of VPN:**

A Virtual Private Network (VPN) is a private connection between two systems or networks built over a shared or public network (usually the Internet). It extends a private network securely across a public infrastructure using encryption and tunneling protocols.

**How VPN Works (Diagram & Explanation):**

(Source: VPN Tunneling Diagram)

VPN works through **tunneling** and **encryption**:

1. Data payload is encrypted at the sending VPN gateway/client.
2. The encrypted packet is encapsulated inside an outer IP packet with public source and destination header addresses.
3. The packet travels securely through the public Internet (the "tunnel").
4. At the receiving VPN gateway, the outer header is stripped and the payload is decrypted to enter the remote network safely.

**Advantages of Using VPN:**

- **Cost Reduction:** Eliminates the need for expensive leased physical lines to connect remote offices.
- **Security & Privacy:** Protects sensitive company data over public networks using robust encryption.

- **Flexibility & Mobility:** Enables remote workers and traveling employees to securely access corporate resources from anywhere.

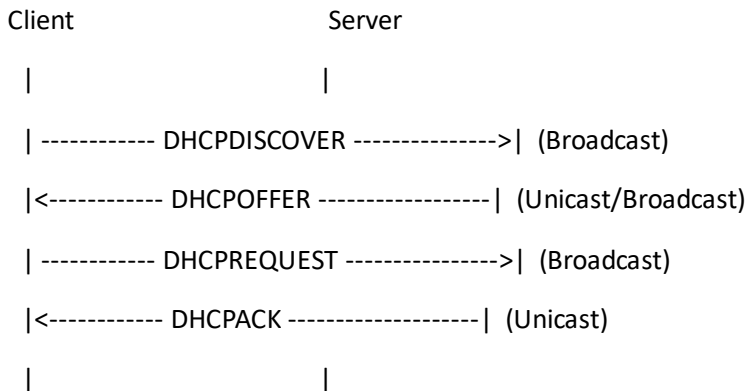
### 7. a. Explain how DHCP Protocol works? [4]

**Answer:**

#### **Purpose of DHCP:**

Dynamic Host Configuration Protocol (DHCP) automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to hosts on a network.

#### **How DHCP Works (DORA Process):**



*(Source: DHCP Client-Server Message Exchange)*

1. **DHCPDISCOVER:** The client boots up and broadcasts a DHCPDISCOVER packet on UDP port 67 searching for available DHCP servers.
2. **DHCPOFFER:** The DHCP server receives the discovery message and responds with a DHCPOFFER packet offering an available IP address, lease time, subnet mask, and gateway settings.
3. **DHCPREQUEST:** The client accepts the offer and broadcasts a DHCPREQUEST packet back to confirm the chosen IP address to all listening servers.
4. **DHCPACK:** The server confirms the assignment by sending a DHCPACK packet, binding the IP address to the host's MAC address for the agreed lease duration.

---

### 7. b. Critically evaluate the role of Network Address Translation (NAT) in modern IP networking.

**Explain how NAT operates within a network infrastructure, and assess its impact on address conservation, security, and end-to-end connectivity. Support your answer with examples of NAT types and their practical applications. [4]**

**Answer:**

#### **Operation of NAT in Network Infrastructure:**

Network Address Translation (NAT) operates on a border router connecting a private internal network to the public Internet. It re-writes the IP header source or destination addresses (and transport layer port numbers) of packets passing through it, maintaining an internal translation table to track active sessions.

### Critical Assessment of Impact:

- **Address Conservation:** NAT prolonged the lifespan of IPv4 significantly by allowing thousands of devices on private IP ranges (e.g., 192.168.x.x or 10.x.x.x) to share a single or small pool of public IP addresses.
- **Security Benefits:** Acts as an implicit stateful firewall. Unsolicited inbound connections from the Internet are blocked because incoming packets do not have a pre-existing entry in the NAT translation table.
- **End-to-End Connectivity Loss:** Breaks the fundamental end-to-end architectural principle of the Internet. Applications that require direct peer-to-peer connections (such as VoIP, P2P file sharing, and online gaming) fail without complex NAT traversal techniques like STUN/TURN or UPnP.

### NAT Types & Practical Applications:

| NAT Type                                      | Mechanism                                                                  | Practical Application                                              |
|-----------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|
| Static NAT                                    | Fixed 1-to-1 mapping of a private IP to a public IP.                       | Hosting public Web/Email servers located inside a private network. |
| Dynamic NAT                                   | Unmapped private IPs grab an available public IP from a pool on demand.    | Enterprise networks with an allocated pool of public IPs.          |
| PAT (Port Address Translation / NAT Overload) | Maps multiple private IPs to 1 public IP using unique source port numbers. | Standard home Wi-Fi routers and small office broadband setups.     |

7. c. Compare and contrast the operational mechanisms of BGP and OSPF in terms of scalability, convergence behavior, and administrative control. Analyze why BGP is preferred in inter-domain routing scenarios, and evaluate its impact on global Internet stability. [4]

Answer:

### Comparison Table:

| Feature                | OSPF (Open Shortest Path First)                                     | BGP (Border Gateway Protocol)                                     |
|------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------|
| Protocol Category      | Interior Gateway Protocol (IGP).                                    | Exterior Gateway Protocol (EGP).                                  |
| Routing Metric / Basis | Link-State (Dijkstra's shortest path based on link cost/bandwidth). | Path-Vector (AS-Path attributes and administrative policy rules). |

|                               |                                                                                                           |                                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Scalability</b>            | Designed for a single Autonomous System (AS). Scales up to thousands of routers using hierarchical areas. | Designed for global scale. Handles the entire global Internet routing table (800,000+ routes).   |
| <b>Convergence Behavior</b>   | <b>Fast convergence.</b> Immediately floods Link State Packets (LSPs) on topology changes.                | <b>Slow convergence.</b> Uses rate-limiters (MRAT timers) to prevent route flap oscillations.    |
| <b>Administrative Control</b> | Technical shortest-path metric; limited policy manipulation.                                              | <b>High policy control.</b> Operators define complex routing rules based on business agreements. |

#### Why BGP is Preferred for Inter-Domain Routing:

1. **Policy-Based Routing:** Inter-domain routing is driven by business relationships (peering, transit, customer agreements) rather than shortest geographical or technical path. BGP allows granular policy rules using attributes like AS-PATH, LOCAL\_PREF, and MED.
2. **Global Scalability:** BGP does not require routers to store full link-state details of foreign networks, storing only path-vector reachability information.

#### Impact on Global Internet Stability:

- **Positive:** Path-vector routing prevents loops across Autonomous Systems, and BGP route dampening suppresses flapping links to protect global internet stability.
- **Negative:** BGP lacks inherent authentication (vulnerable to BGP hijacking and misconfigurations), where incorrect route advertisements can cause global outages or traffic interception.